Smart VMS — System Architecture Submission

Aligned with MoICT&NG Prototype Showcase architecture guidance

Smart VMS (Smart Visitor Management System) — Government Systems Prototype Showcase and National Innovator Registry

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Showcase repo	https://github.com/paradigmaRnD/smartvms-showcase

1. Purpose of This Document

This submission addresses the Ministry of ICT and National Guidance architecture elements: components, communication, data storage and security, user access, integrations, hosting, security controls, and data flows.

Smart VMS follows a layered architecture: client applications (mobile and web) communicate with application services over encrypted channels. Business rules enforce visitor policies, notifications, and reporting. Data is persisted in segregated stores for operational records and administration. The design supports scaling across multiple application instances and is structured for interoperability with external services through documented APIs.

2. Architecture Diagram (Logical)

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+------------------------------------------------------------------+ | USERS (Staff &
Administrators) | +------------------------------------------------------------------+ | | v v
+----------------------+ +----------------------+ | MOBILE CLIENT | | WEB ADMIN PORTAL | |
Reception / Security | | Configuration | +----------+-----------+ +----------+-----------+ | |
+----------------+------------------+ | HTTPS / TLS 1.2+ v
+------------------------------------------------------------------+ | APPLICATION SERVICES
(Node.js) | | Visitor workflows | Policies | Notifications | Audit hooks |
+------------------------------------------------------------------+ |
+-------------+-------------+ v v v +---------------+ +---------+ +--------------+ | Identity &
| | Primary | | Admin / | | Sessions | | Data | | Media Store| +---------------+ +---------+
+--------------+ | +-------------+-------------+ v v +---------------+ +---------------+ | Email
(SMTP) | | Future Govt | | notifications | | APIs (planned)| +---------------+ +---------------+
```

3. Main Components

Component	Role	Technology (summary)
Mobile Client	Reception and security staff: visitor registration, check-in/out, premises validation, limited offline operation	Cross-platform mobile framework
Web Administration Portal	Organisation administrators: user management, visitor records, configuration, reports	Modern web application (browser-based)
Application Services	Core business logic: visits, premises, policies, subscriptions, incidents, notifications	Node.js modular services
Identity & Session Service	Authentication, session management, role enforcement	Industry-standard token-based access
Primary Data Store	Operational visitor and visit records, organisation configuration	Document-oriented database
Administrative Data Store	Supplementary relational records for web administration module	Relational database
Media & Document Storage	Visitor-related attachments (e.g. capture images) with access controls	Secured object storage behind application APIs

4. MoICT&NG Requested Elements

The table below maps directly to the Ministry guidance checklist (Frontend, Backend, Database, Authentication, APIs, Hosting, Security, Data Flow).

Area	Description
Frontend / User Interface	Mobile app for reception and security (check-in, check-out, premises validation). Web portal for administrators (records, users, configuration, reports). Primary workflows: register visitor → validate access → notify host → check-out → audit/report.
Backend / Application Layer	Node.js application services host business logic. Requests authenticated and authorised per role. Stateless service instances suitable for load balancing. Request validation and audit hooks on sensitive operations.

Area	Description
Database Layer	Categories stored: organisation profile, users and roles, premises, visitor profiles, visit events (check-in/out timestamps), audit logs, configuration. Primary operational store plus relational store for admin module. Hosting: cloud VPS (production); on-premise deployment path available. Encryption at rest on infrastructure; access restricted to application layer.
Authentication and Access	Credential-based login with secure password handling. Session tokens validated on each request. Roles include Administrator, Manager, Receptionist, Security, and Host with least-privilege permissions. Organisation-level data isolation. MFA: planned; not required for initial deployment.
APIs and Integrations	Exposes REST APIs for visitor lifecycle and administration (consumption by mobile and web clients). Consumes: email (SMTP) for notifications; optional calendar connectivity. Planned: National Identification and other Government platforms via standard integration interfaces. No live NIRA/URA/NITA-U connection in current production.
Hosting and Infrastructure	Production: cloud virtual private server with HTTPS termination (TLS 1.2+). Reverse proxy in front of application services. Hybrid/on-premise deployment supported for institutions requiring local hosting. Geographic hosting: contractor-operated cloud; migration to NITA-U-aligned hosting under roadmap.
Security Controls	TLS for all client–server traffic. Encryption at rest on database and file storage. RBAC enforced in application layer. Comprehensive audit logging (user, action, timestamp, source). Input validation and upload controls on media. Aligned with Uganda Data Protection and Privacy Act, 2019.
Data Flow	Data enters at mobile/web clients → TLS → application services → authorisation check → persistence → optional notification to host. Data exits via authorised reports/exports and audit review. No public anonymous access to visitor records.

5. Data Flows

- Check-in: Staff client → HTTPS → Application services → Role validation → Visit record created → Host notification → Audit entry
- Check-out: Staff client → HTTPS → Visit record updated → Audit entry
- Administration: Web portal → HTTPS → Authenticated API → Configuration or report → Audit entry
- Premises validation: Client presents access token → Server validates policy and timing → Visit associated with premises
- Offline sync: Mobile queues events locally → On connectivity, encrypted synchronisation with conflict handling

6. User Access

- Android mobile application (reception and security)
- Web browser (organisation administrators)
- REST APIs (authenticated integrations — documentation in showcase repository)

7. Integrations

Current

- Email (SMTP) for account verification and visit notifications
- Optional calendar integration for scheduling-related features

Planned

- National Identification / digital identity services
- Government integration layer and sector platforms as standards are adopted
- NITA-U aligned hosting where required by procurement policy

8. Security Controls

- Encrypted transport (TLS) on all client connections
- Session-based authentication with per-request validation
- Role-based access control with organisation isolation
- Audit trail for administrative and visitor-data actions
- Secure handling of uploaded media; metadata minimisation
- Rate limiting and validation on public-facing API endpoints

9. Scalability & Resilience

Application tier designed stateless for horizontal scaling behind a load balancer. Database tier supports replication. Suitable for institutional deployments from single premises to multi-site organisations.

10. Confidentiality

Submitted for MoICT&NG evaluation and National Innovator Registry profiling only. Confidential per application terms. Does not transfer intellectual property. Proprietary implementation details, credentials, and algorithms are not included in this public summary or the showcase repository.

Assessment repository: https://github.com/paradigmaRnD/smartvms-showcase